



internal nodes and outcomes as branches.

4. **Knowledge-based systems.** These are programs that represent knowledge and solve complex problems by reasoning out how knowledge artefacts are related or not related. These are used to show relationships between features in fashion styling.

5. **Genetic algorithms.** These are search techniques that look for approximate or exact solutions to optimisation problems. They are guided by a fitness function.

Smart Fashion: A simple AI fashion styling system

Generally, fashion styling involves the following seven steps: Choosing a theme, selecting a primary colour, mixing and matching clothing pieces, selecting accessories, model or client fitting, hair styling and makeup. AI computer modelling program usually includes the following functions: Representation of the garments computationally (focuses on fashion object), detection, tracking and forecasting fashion trends (about fashion process) and modelling human stylist behaviour (a mixture of both).

Smart Fashion is a machine learning application that recommends fashion looks by learning user preferences through Multilayer Perceptron model. The system scores users' customised style looks based on fashion trends and users' personal style history.

Smart Fashion system comprises five major components for fashion styling task: Initialisation program, database, smart engine, learning components and user interface. It uses artificial neural networks to learn user preferences. It creates a database containing, say, 32 dresses and 20 shoes for four different events, encodes a standard style rules engine, generates more than 500 looks and ranks them by a final score in descending order. The score indicates how fashionable each look is based on users' feedback.

The learning component trains artificial neural network to learn users' personal preferences and adjusts the final score. The system allows users to customise a fashion look and then evaluate it. This feature provides a shopping guide to inform users' purchase plans.

Smart Fashion provides a numeric



Fig. 7: Fabric flowers dress embedded with LEDs under the Marchesa label that changes colours (Source: <https://qz.com>)

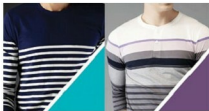


Fig. 8: Moda Rapido t-shirts from Mynta designed using AI (Source: <http://blog.myntra.com>)

evaluation for user preferences. The system is initialised by the user taking a fashion personality questionnaire. The output of the questionnaire is stored in the system. So users can restart the program by clicking Reset button, avoiding the need to retake the quiz. The questionnaire is described in the box below.

For each style, the system has a set of default database and styling rules. Using style rules, the Smart Engine pairs up dress and shoes, assigns initial scores to each pair and creates a 'pairs' table in the database. Two main outputs of the system are: A recommender system to recommend users' fashion looks based on user interactions and an evaluation system to evaluate users' customised looks with a score that assists users' shopping plans.

System learning predicts a pair's score based on correlations between scores and item attributes. The dataset is prepared and pre-processed. For example, all the dresses and shoes are collected in typical standard style from various sources such as websites and designers' collections. All clothing items and shoe items are assigned a number of attributes, which are compiled in tables. Standard rules are created for the database. Every rule interprets a binary (positive or negative) relationship between pairs of attributes. Boredom, one of the major

factors in fashion model, is created in fashion styling for wearing the same looks several times. So, a count of 'wear' is added in the program as a parameter of boredom.

The output of the Smart Engine is a pair table for each event, which stores the pair data and score, and pair's views, which stores 'like' and 'dislike' data. The data for 'like' and 'dislike' is created based on the learning components and evaluation based on the selected attributes. The learning task is completed using a number of Machine Learning algorithms and data preprocessing to predict the final score based on correlation with item

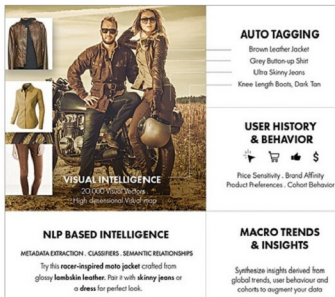


Fig. 9: Visual recognition, user behaviour, NLP and brand data as inputs to analyse and forecast (Source: <https://vue.ai>)